

Enhancing Healthcare Resource Allocation Through Large Language Models

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Abstract: Recognizing the growing capabilities of large language models (LLMs) and their potential in healthcare, this study explores the application of LLMs in healthcare resource allocation using Prompt Engineering, Retrieval-Augmented Generation (RAG), and Tool Utilization. It addresses both optimizable and non-optimizable challenges in allocating operating rooms (ORs), postoperative beds, and surgeons, while also identifying key factors like ethical and legal constraints through a medical knowledge Q&A survey. Among the seven evaluated LLMs, including LaMDA 2, PaLM 2, and Qwen, ChatGPT-4o demonstrated superior performance by reducing OR and surgeon overtime, alleviating peak bed demand, and achieving the highest accuracy in medical knowledge queries. Comprehensive comparisons with traditional methods (exact and heuristic algorithm), varying problem sizes, and hybrid approaches from the literature revealed that as problem size increased, LLMs performed better and faster by integrating historical experience with new data. They adapted to changes in problem scale or demand without requiring re-optimization, effectively addressing the runtime limitations of traditional methods. These findings underscore the potential of LLMs in advancing dynamic and efficient healthcare resource management.

Keywords: Large language models, Combinatorial optimization, Surgery Scheduling, Medical Q&A.

1. Introduction

Healthcare resource allocation has always been a critical issue due to the limited availability of operating rooms (ORs), recovery beds, and healthcare staff. The COVID-19 pandemic further exposed the fragility of healthcare systems, highlighting the need for more efficient resource allocation strategies. Recent advancements in artificial intelligence, particularly large language models (LLMs), offer promising avenues for addressing these challenges by enabling better predictions, optimizations, and decision-making.

LLMs such as GPT and their successors, have become transformative tools across various fields, showing their exceptional ability to process, analyze, and generate human-like text [1-3]. These models, trained on vast amounts of data, excel at understanding context, recognizing patterns, and offering data-driven solutions. Their applications have extended beyond simple text generation to complex problem-solving tasks, including optimization, prediction, and decision support [4],[5]. In healthcare, where data is both vast and often unstructured, LLMs offer a unique

opportunity to streamline operations, enhance decision-making, and ultimately improve patient care outcomes [6],[7].

This research focuses on optimizing resource allocation in hospitals, particularly the scheduling of operating rooms, the allocation of postoperative beds, and the assignment of healthcare staff. Poor surgery scheduling, improper bed allocation, or staff shortages can lead to delays, increased operational costs, and compromised patient care [8]. Traditionally, resource allocation in healthcare has relied on heuristic methods or statistical models, which often require substantial manpower and resources [9]. As the demand for more personalized and efficient healthcare grows, LLMs are poised to play a transformative role in the sector [10],[11].

The core problem addressed in this study is the allocation of healthcare resources in a hospital setting. This includes minimizing OR costs (opening and overtime fees), reducing surgeon overtime, and managing peak bed demand. The complexity arises from multi-objective optimization, as well as the dynamic and stochastic nature of patient arrivals and resource availability.

One of the key advantages of LLMs lies in their ability to integrate vast amounts of diverse information, including real-time data, medical guidelines, and patient feedback, to provide tailored and adaptive solutions [12],[13]. In the context of healthcare, LLMs can be used not only to support scheduling and allocation decisions but also to predict future demand for resources, simulate different scenarios, and assess potential outcomes. This predictive capability is particularly valuable in managing uncertainties that often arise in healthcare settings, such as fluctuations in patient arrivals, unexpected staff shortages, or shifts in resource demand due to external events [14].

Moreover, LLMs can assist in bridging the gap between various stakeholders in the healthcare system. By effectively summarizing and interpreting complex data, LLMs can facilitate communication among healthcare providers, administrators, and patients [15]. For instance, LLMs can help translate intricate scheduling data into understandable insights for hospital staff or patients, ensuring that all parties are well-informed and engaged in the care process. This enhanced communication can lead to better alignment between hospital resources and patient needs, ultimately improving the overall quality of care [16].

In this study, we leverage LLMs to generate optimized solutions and adapt dynamically to changes in problem scale and resource demand. Compared to traditional exact and heuristic algorithms, LLMs demonstrate remarkable capabilities in efficiently addressing multi-objective problems. Our main contributions are as follows:

1. We introduce an LLM-based framework to address both optimizable and non-optimizable aspects of medical resource allocation, reducing hospital costs, improving resource efficiency, and enhancing satisfaction for surgeons and patients.
2. Using Prompt Engineering, RAG, and Tool Utilization, LLMs integrate medical data and adapt to changes in problem scale or demand without re-optimization, addressing runtime limitations of traditional methods and advancing dynamic resource allocation.
3. We conduct comprehensive experiments on different LLMs performance, comparisons with traditional methods (exact and heuristic), scalability across problem sizes, and hybrid methods from recent literature. The results demonstrate that LLMs effectively solve multi-objective medical resource allocation.

4. We identify key factors, including ethical and legal constraints, influencing medical resource allocation through a comprehensive medical knowledge Q&A questionnaire.

The rest of the paper is organized as follows: Section 2 reviews healthcare resource allocation models, algorithms, and LLM applications in healthcare. Section 3 introduces the mathematical model for resource allocation, while Section 4 explains the LLM framework for solving these problems. Section 5 covers the optimizable aspects and comparative experiments, and Section 6 focuses on the non-optimizable aspects and results. Finally, Section 7 concludes the paper.

2. Literature review

2.1. Algorithms for Medical Resource Allocation

The allocation of medical resources, particularly in complex environments like hospitals, has been extensively studied. Various algorithmic approaches have been proposed to address this issue, ranging from exact algorithms to intelligent optimization techniques.

Exact Algorithms are often used in solving resource allocation problems with precise calculations, aiming to find the optimal solution. However, they are computationally expensive and struggle with large-scale, real-time data, especially in dynamic environments like healthcare, where patient volumes and resource availability fluctuate [17]. For example, column generation algorithms, benders decomposition, etc., but such models require extensive computational resources and often fail to adapt quickly to changing circumstances in real-world healthcare settings [18],[19].

Heuristic and Metaheuristic Algorithms provide alternative solutions by approximating optimal results with less computational effort. Techniques such as Genetic Algorithms, Simulated Annealing, and Particle Swarm Optimization have been used to address complex scheduling problems [20-23]. Although they offer more flexibility compared to exact algorithms, these approaches often require careful tuning of parameters and are sensitive to problem-specific characteristics [24]. For instance, the performance of these algorithms can vary based on how the problem is encoded, making them less adaptable to the dynamic and uncertain nature of healthcare environments.

Several studies [25],[26] have explored multi-objective optimization algorithms in healthcare resource allocation. While these studies are significant in addressing healthcare scheduling issues, they rely on traditional optimization techniques such as evolutionary algorithms and stochastic simulations, which can be computationally expensive and less adaptable to dynamic changes. In contrast, our proposed approach uses LLMs to provide faster and more accurate solutions, offering a novel and more efficient approach for multi-objective optimization in healthcare resource allocation.

As the need for more flexible and adaptive solutions grows, researchers have begun exploring hybrid models that combine the strengths of various optimization techniques in healthcare resource allocation. For example, Genetic Algorithms have been combined with machine learning models to optimize staffing schedules in hospitals [27]. Another hybrid approach combines Simulated Annealing with Reinforcement Learning for dynamic nurse rostering [28]. These hybrid methods can enhance the performance of traditional optimization techniques by improving adaptability to fluctuating healthcare demands.

Despite these advancements, there is still a demand for more efficient, less resource-intensive approaches that can handle the high complexity and uncertainty of healthcare environments. The integration of multiple algorithms often requires substantial computational resources and large datasets, making it difficult to scale effectively in real-time. This has led to the exploration of LLMs as a potential solution.

2.2. Large Language Models in Healthcare

LLMs have gained significant attention for their ability to process vast amounts of unstructured data, making them particularly well-suited for healthcare applications. These models have been used to address a range of tasks, including prediction, optimization, and decision-making, with notable success in improving efficiency and accuracy in clinical settings.

Prediction and Diagnosis: LLMs have been applied in predictive modeling for patient outcomes, such as surgery durations, recovery times, and disease progression. Studies like Rajpurkar & Lungren [29] have explored the use of AI in medical image interpretation, while Haider et al. [30] focused on LLMs for breast cancer classification. LLMs can rapidly analyze medical images and patient records to provide diagnostic support, improving decision-making in critical scenarios.

Optimization in Resource Allocation: LLMs are also proving to be effective in optimizing resource allocation in healthcare settings. Liu et al. [31] demonstrated the parameter-efficient fine-tuning of LLMs for multi-task medical applications, highlighting their ability to optimize various resource-intensive processes.

Medical Decision Support: LLMs have been employed to generate structured medical reports, such as in Li et al. [32], where LLMs were used to create chest X-ray reports. Furthermore, Zhou et al. [33] applied LLMs for biomedical relation extraction, enabling the models to identify relationships between medical entities and improve the interpretation of clinical data.

Medical Questionnaires and Patient Summaries: LLMs also show promise in handling survey-related tasks, such as answering medical questionnaires and generating patient summaries. Research by Jin et al. [34] developed a large-scale dataset from medical exams, demonstrating how LLMs can effectively respond to open-domain medical questions. Similarly, Longwell et al. [35] evaluated LLM performance on oncology exam questions, revealing their potential to streamline medical education and patient interaction through question answering systems.

Building on these advancements, this study leverages LLMs to tackle healthcare resource allocation challenges, optimizing OR scheduling, postoperative bed allocation, and staff assignment for more efficient resource management. Additionally, LLMs analyze medical surveys, offering insights into patient satisfaction and staff well-being to guide allocation strategies.

3. Mathematical Model of Medical Resource Allocation

The model mainly optimizes the allocation of OR and postoperative beds for elective surgery patients. The objectives include minimizing the total cost about ORs (including the opening fees and overtime fees), reducing the overtime hours for surgeons, and decreasing peak demand for beds. This helps stagger patient appointments and reduce idle bed time. The parameters, variables, and mathematical formulas are described below:

1.Parameters and Variables:

x_{ijk} : Decision variable, indicating whether surgery i is assigned to OR j on day t and by surgeon k (1 if assigned, 0 otherwise)

Y_{jt} : Decision variable, indicating whether OR j is open on day t (1 if assigned, 0 otherwise)

R and R_{max} : Regular and Maximum working hours per surgeon,

Q and Q_{max} : Regular and Maximum opening hours per OR,

a : Hourly overtime cost per OR,

C : The opening costs per OR,

D_i : Surgery duration for elective patient i ,

Los_i : length of stay (LOS) of elective patient i ,

2.Objective Functions

f1: This objective seeks to minimize the overall OR expenses, which include two components, opening costs: the fixed costs associated with keeping the ORs operational during the day; overtime costs: the additional costs incurred when surgeries extend beyond the designated operational hours, requiring the OR to remain open longer, as shows in Eq. (1).

$$\min f1 = a * \sum_{j=1}^J \sum_{t=1}^T O_{jt} + C * \sum_{j=1}^J \sum_{t=1}^T Y_{jt} \quad (1)$$

f2: Eq. (2) aims to reduce the total overtime hours accumulated by surgeons when surgeries extend beyond regular working hours. The model ensures better workload management, prevents burnout.

$$\min f2 = \sum_{t=1}^T \sum_{k=1}^K \left(\max(0, \sum_{i=1}^I \sum_{j=1}^J x_{ijk} * D_i - R) \right) \quad (2)$$

f3: Eq. (3) focuses on reducing the maximum number of recovery beds needed on any given day. By minimizing peak bed demand, the model help prevent overcrowding and allowing for smoother patient flow and resource allocation, where B_t is the daily demand for beds.

$$\min f3 = \max(B_1, B_2, \dots, B_T) \quad (3)$$

3. Constraints

Constraint (4) defines the non-negative variable for the total overtime incurred by OR j on day t . This ensures that only excess hours beyond the regular schedule are accounted for, avoiding any negative values, Where O_{jt} is Overtime in OR j on day t .

$$O_{jt} = \max\{0, \sum_{k=1}^K \sum_{i=1}^I D_i * x_{ijk} - Q\}, \forall j, \forall t \quad (4)$$

Constraints (5) ensure that the opening time of the OR does not exceed the maximum allowed limit. This prevents the OR from being scheduled beyond its designated operational hours.

$$\sum_{k=1}^K \sum_{i=1}^I D_i * x_{ijk} \leq Qmax, \forall j, \forall t \quad (5)$$

Constraints (6) ensure that the working time of the surgeon does not exceed the maximum allowable limit. This helps to prevent overworking the surgeon and ensures compliance with work-hour regulations.

$$\sum_{j=1}^J \sum_{i=1}^I D_i * x_{ijt} \leq Rmax, \forall k, \forall t \quad (6)$$

Constraint (7) refers to the bed capacity. Constraint (4) refers to the bed capacity, ensuring that the number of patients assigned to recovery beds does not exceed the available number of beds at any given time.

$$\sum_{i=1}^I \sum_{o=t-LoS_i+1}^t \sum_{k=1}^K \sum_{j=1}^J x_{ijk} \leq B_t, \forall t \quad (7)$$

Constraint (8) stipulates that each patient is assigned to a surgery only once. This ensures that there are no duplicate assignments, and each patient is properly scheduled and allocated to the necessary resources for their procedure.

$$\sum_{t=1}^T \sum_{j=1}^J \sum_{k=1}^K x_{ijk} \leq 1, \forall i \quad (8)$$

Constraint (9) represents the relationship between two decision variables. Constraint (6) represents the relationship between two decision variables, aligning OR usage with surgeon availability, to ensure efficient resource utilization and logical decision-making within the model.

$$x_{ijkt} \leq Y_{jt}, \forall t, \forall j, \forall k, \forall i \quad (9)$$

4. LLMs for Medical Resource Allocation

When using LLMs to solve medical resource allocation problems, there are three related techniques that are particularly useful: Prompt Engineering, Retrieval-Augmented Generation (RAG), and Tool Utilization. These three techniques are explained in detail, along with their application in medical resource allocation:

4.1.1. Prompt Engineering

Prompt engineering is the process of optimizing the interaction with LLMs to ensure high-quality, relevant, and practical output. By providing specific prompts, the model can offer optimization suggestions for tasks. Pseudocode 1 represents prompt engineering for solving medical resource allocation.

Pseudocode 1: Prompt engineering for solving medical resource allocation	
1.	Step 1: Problem Understanding and Contextual Analysis
2.	Example Task: "Understand the context of hospital resource allocation, including the roles
3.	of elective patients, operating rooms, and postoperative beds."
4.	Step 2: Construct Prompt - Allocate Operating Rooms and Postoperative Recovery Beds
5.	Example Prompt: "Allocate operating rooms and postoperative recovery beds to elective
6.	patients to minimize resource strain."
7.	Step 3: Construct Prompt - Explain Constraints and Objective Functions
8.	Example Prompt: "Such as minimizing overtime and balancing bed demand."
9.	Step 4: Input - Patient and Resource Data Collection
10.	Example Input: "Input a dataset of 50 elective patients, including surgery duration, length
11.	of stay, and assigned surgeons."
12.	Step 5: Data Preprocessing and Normalization
13.	Example Task: "Normalize surgery duration, length of stay to ensure consistent data input."
15.	Step 6: LLM-Generated Solution and Decision Explanation
16.	Example Output: "Assign patient A to operating room 1 on day 1, ensuring bed availability
17.	for 5 days. Explanation: This minimizes overtime for both OR and surgeon."

4.1.2. Retrieval-Augmented Generation (RAG)

RAG is a technique that combines retrieval and generation. It first retrieves relevant information from a knowledge base and then generates the corresponding output using LLMs. This approach has the advantage of using specific context and the latest information when generating responses. In medical resource allocation, applications of RAG include:

a) Data Usage: RAG loads data, including patient information and the availability of medical resources such as OR availability and bed occupancy. This ensures that the generated solutions are grounded in accurate and up-to-date information.

b) Dynamic Adaptation to Changes: RAG enables LLMs to dynamically adapt to changes in problem scale or resource demand by integrating historical experience with new data, without requiring complete re-optimization.

4.1.3. Tool Utilization

Tool utilization involves combining LLMs with other software tools to solve more complex tasks. For example, integrating the model with data analysis, optimization algorithms, or plotting tools. Specific applications in medical resource allocation include:

a) Integration with Tools: Combining LLM with existing tools to automatically generate and solve optimization models for surgery scheduling, bed allocation, etc.

b) Results Visualization: Using plotting tools to generate visual outputs, such as bar charts

showing OR availability.

4.1.4. Process framework for LLMs

In managing the allocation of ORs, postoperative beds, and healthcare staff, prompt engineering can be used to define specific resource allocation problems; RAG can be used to retrieve real-time occupancy status and patient data from hospital information systems to generate optimization suggestions; and tool utilization can be used to complete complex calculations and provide intuitive analysis results. The process for solving healthcare resource allocation using LLMs is outlined in Pseudocode 2, and the framework is illustrated in Fig 1.

Pseudocode 2: The framework where LLMs solve medical resource allocation	
1.	Step 1. Initialize Problem Parameters:
2.	- Define patient data: surgery duration, LOS, and surgeon availability.
3.	- Set resource constraints: OR capacity, bed limits, and working hours.
4.	Step 2. Apply Prompt Engineering:
5.	- Construct task-specific prompts for optimization goals
6.	- Include constraints and input data in the prompt.
7.	Step 3. Execute Retrieval-Augmented Generation:
8.	- Load hospital data (e.g., OR occupancy, surgery queue).
9.	- Retrieve changes in patient information or resource demands
10.	Step 4. Utilize Tool Integration:
11.	- Pass input and retrieved data to optimization solvers (e.g., MATLAB models).
12.	- Generate and visualize intermediate solutions (e.g., Gantt charts for OR).
16.	Step 5. Output Solution:
17.	- Display the final resource allocation plan in a user-friendly format.
18.	- Provide detailed explanations for the decisions (e.g., "Assign Patient A to OR 1 on Day 2 to minimize bed demand").
19.	Step 6. Handle Changes in Data or Resource Demands:
20.	- Load new data and integrate it with historical insights.
21.	- Dynamically adjust constraints or priorities based on updated requirements
22.	Step 7. Finalize and Export:
23.	- Save the allocation plan and results.
24.	- Present visual summaries (e.g., OR schedules, bed utilization trends).

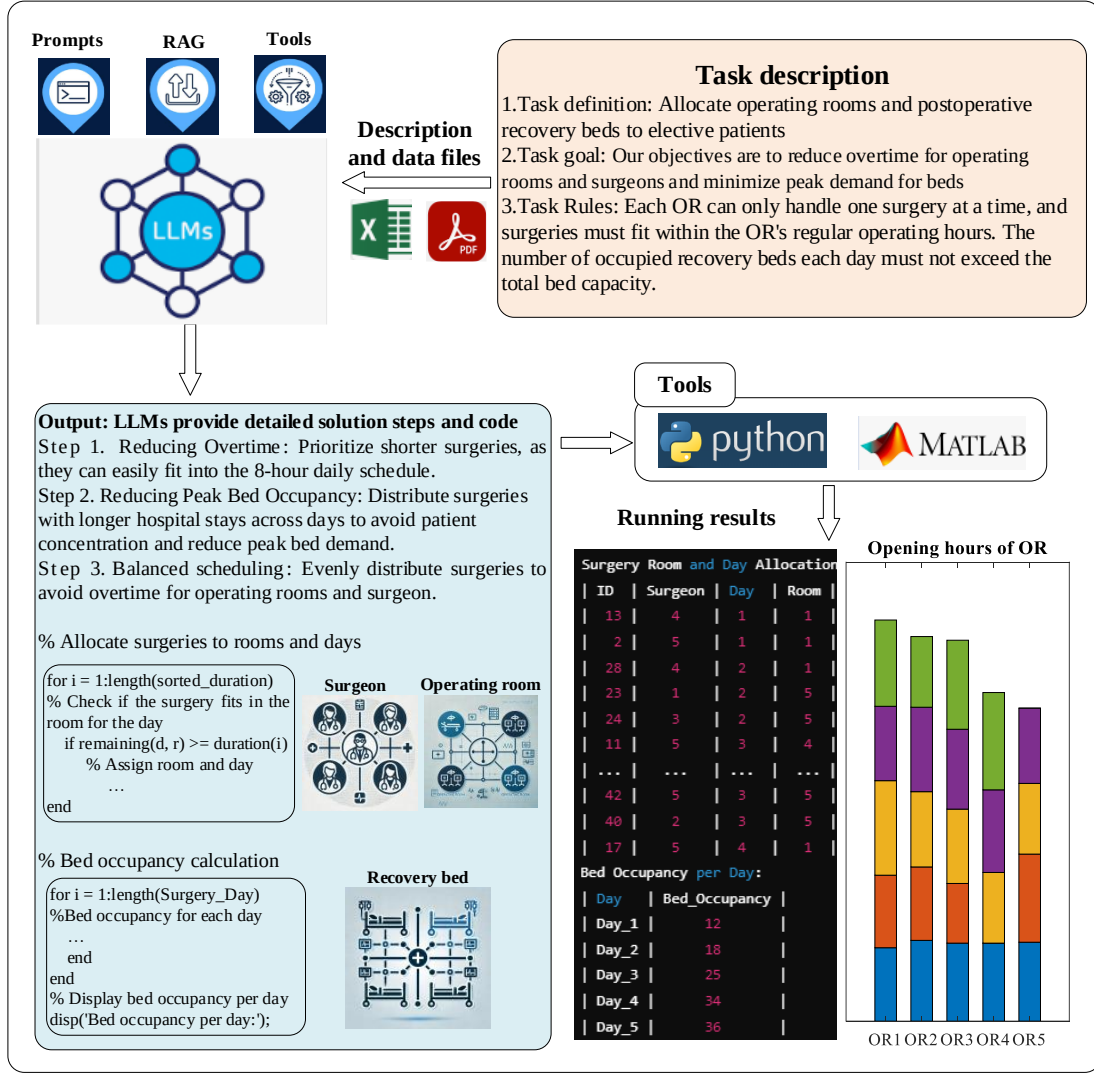


Fig. 1. The framework where LLMs solve medical resource allocation.

5. Optimizable aspects and comparative experiments

The practical application involves the allocation of ORs and postoperative beds for elective patients at a hospital in Lyon, France, over one week period. As shown in Table 1, there are a total of 50 elective surgery patients, and each information includes surgery duration (in hours) and their LOS (in days). Table 2 displays the parameter settings of model, including the number of ORs, the planning period, etc.

Table 1. The information for elective patients

No.	Duration	Los	No.	Duration	Los	No.	Duration	Los
1	5.6	6	11	4.7	3	21	3.2	2
2	2.3	3	12	5.7	8	22	4.4	6
3	6.3	7	13	2.2	7	23	5	8
4	5.2	6	14	5.8	6	24	4.6	6
5	3	6	15	2.9	3	25	3.5	6
6	2.5	7	16	6.5	6	26	6.4	7
7	4.8	5	17	2.9	8	27	4.6	4

8	3.7	8	18	2.1	3	28	6.2	4
9	1.8	4	19	4.8	5		...	
10	5.2	6	20	1.7	8	50	4.1	3

Table 2. The settings of model parameters

Parameter	Explanation	Value
T	Planning horizon	5
J	Number of OR	5
K	Number of Surgeon	5
Q	Regular opening hours for OR	8
Qmax	Maximum opening hours for OR	10
a	Hourly overtime fee of OR	200
C	Opening fee of OR	1000
R	Regular working hours for surgeon	8
Rmax	Maximum working hours for surgeon	8

5.1. Result analysis

Fig. 2 is a Gantt chart based on a resource allocation plan generated by an LLM: ChatGPT-4o. The numbers [a, b] on the rectangle represent patient a and their attending surgeon b. The color of the rectangle represents the surgeon, the length indicates the surgery duration, the horizontal axis shows the surgery date, and the vertical axis represents the OR.

For each OR or surgeon, any time exceeding 8 hours is considered overtime. Regarding the ORs, two rooms on day 2 and day 5 were not opened, which saved costs and kept them available for emergencies. The total overtime in the open ORs amounted to 13.2 hours, and the total cost, including opening fees and overtime fees, was 25,640(*f1*). The work hours of the five surgeons were evenly distributed over five days, with a total overtime of 14.5 hours (*f2*).

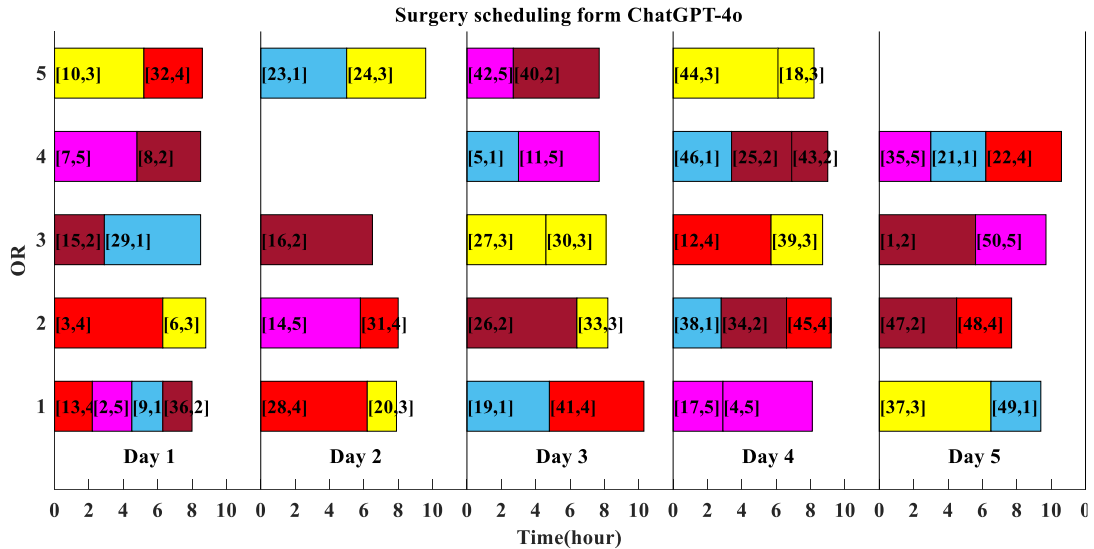


Fig. 2. The Gantt chart of the allocation plan from ChatGPT-4o

Fig. 3 illustrates the OR opening hour, surgeons' working hours, and bed demand. Although some ORs or surgeons worked more than 8 hours on certain days, the total working hours over the five days did not significantly exceed 40 hours, indicating that the scheduling was reasonable and feasible. Regarding bed allocation, the peak demand was 36 beds on Day 5 (*f3*), primarily because the LOS for many patients ranged between 4 and 8 days, leading to an accumulation of bed

demand on the fifth day.

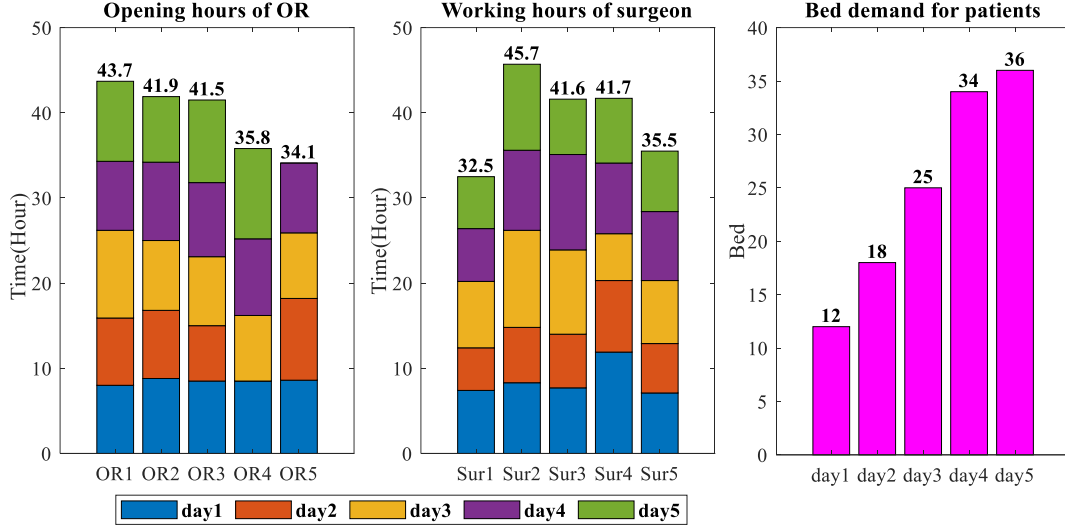


Fig. 3. The working time of OR and surgeons and bed demand

5.2. Experiment comparison

5.2.1. Comparison between LLMs

For comparison, we conducted comparative experiments using several leading LLMs, including ChatGPT-3.5 from OpenAI, LaMDA 2 and PaLM 2 from Google, LLaMA 2 from Meta AI, Qwen from Alibaba, and Ernie from Baidu. Below, each LLM is described in detail. Fig. 4 illustrates the Gantt charts of the allocation results generated by these LLMs.

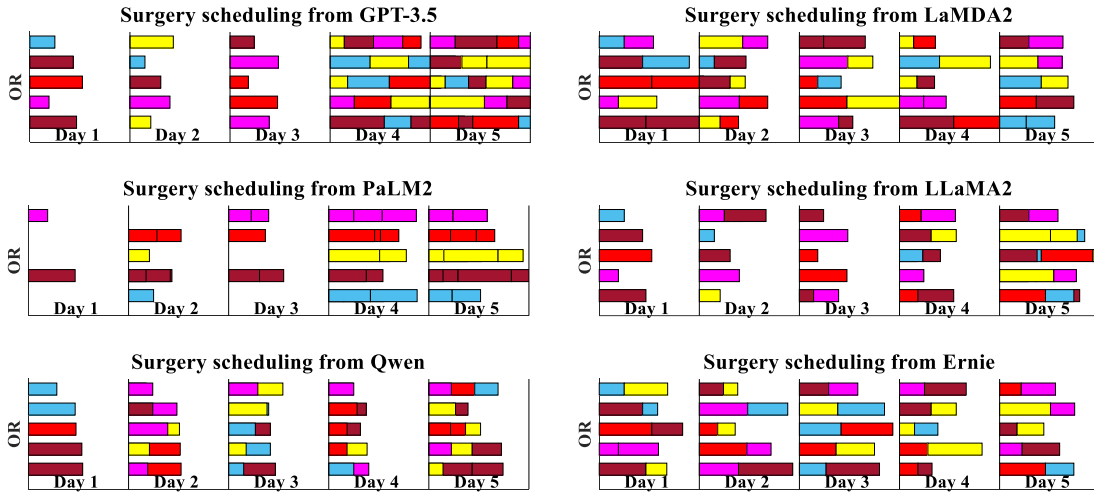


Fig. 4. Gantt chart of the allocation plan of other 6 LLMs

ChatGPT-3.5 and LLaMA 2 assigned only one patient per OR, leading to most patients being concentrated on the fifth day, resulting in an imbalanced utilization of ORs. LaMDA2 and Ernie evenly allocated two patients per OR but lacked sequencing and planning, which caused overtime for both ORs and surgeons. PaLM 2 assigned patients to ORs based on the attending surgeon,

grouping patients with the same surgeon in the same OR. However, PaLM 2 and Qwen only allocated ORs for 38 and 46 patients, respectively, indicating omissions in the assignments. Table 3 presents the three objective functions and runtime for these methods.

Table 3. Objective functions and running times of the LLMs

Model	f1	f2	f3	Time(s)
ChatGPT-3.5	33520	43.8	49	15.3
LaMDA 2	29940	27.6	44	9.6
PaLM 2	26300	36.5	38	8.4
LLaMA 2	31180	45.7	43	12.4
Qwen	27540	37.6	43	10.9
Ernie	29100	33.7	44	9.7
ChatGPT-4o	25640	14.5	36	7.6

Fig. 5 presents the three objective functions, total overtime hours for ORs, and runtime about these LLMs. ChatGPT-4o performs the best on three objective functions and has the shortest response time, which is beneficial for reducing hospital costs, surgeon overtime, and peak bed demand, thereby improving the utilization of medical resources. PaLM 2 also shows relatively small values for the three objective functions, while Qwen has the shortest total overtime. However, both models suffer from data hallucination issues, with PaLM 2 and Qwen failing to allocate ORs for 11 and 6 patients, respectively, making their solutions infeasible.

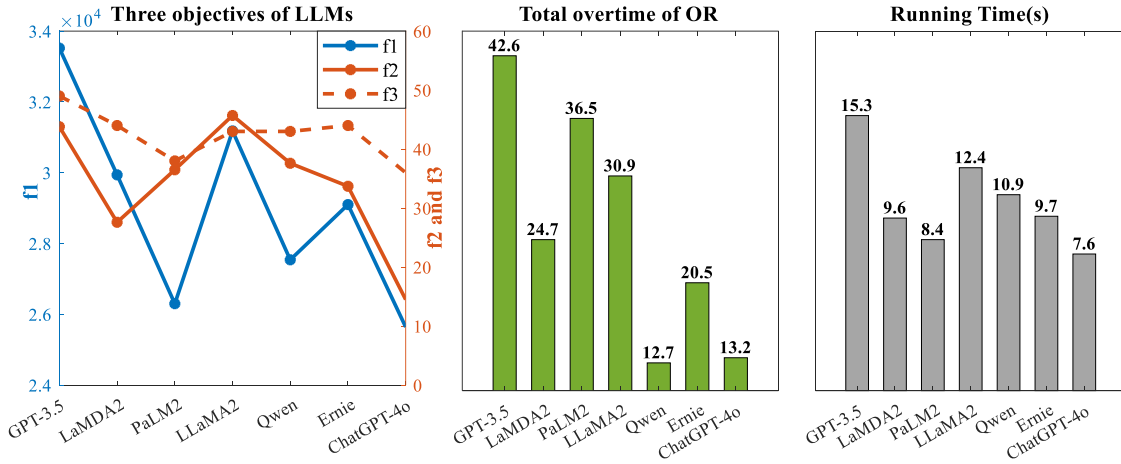


Fig. 5. The three objectives, OR overtime and runtime about these LLMs

5.2.2. Comparison between algorithms

To demonstrate the effectiveness of LLMs in solving multi-objective resource allocation problems, we compared their performance with two traditional methods: an exact algorithm (Column Generation) and a heuristic algorithm (Non-dominated Sorting Genetic Algorithm II).

5.2.2.1 Exact algorithm for Medical Resource Allocation

Column Generation is a decomposition method that iteratively adds variables (“columns”) to the restricted master problem, improving the solution step by step [36],[37]. Multi-objective problems are transformed into a single-objective formulation using a weighted sum approach with

adjustable weights (w_1, w_2, w_3). This method efficiently solves linear programming problems but requires careful balancing of objectives with differing scales. The process for solving healthcare resource allocation using Column Generation is outlined in Pseudocode 3.

Pseudocode 3: Column Generation for Resource Allocation Problem
<pre> 1. % Step 1: Initialization 2. Initialize parameters and decision variables 3. x = zeros (n_surgeries, n_or, n_days, n_surgeons); y = zeros (n_or, n_days); B = zeros (n_days, 1); 4. % Step 2: Define objective functions 5. f1 = Minimize costs; f2 = Minimize surgeon overtime; f3 = Minimize peak bed demand; 6. % Step 3: Combine objectives 7. objective = w1*f1 + w2*f2 + w3*f3; % Define Master Problem (MP) with objective 8. MP.objective = @(x, y, O, B) objective(a, O, C, y, x, D, R, B); 9. % Step 2: Solve Relaxed Master Problem (MP) 10. dual_values = solve_relaxed_MP(MP); % Obtain dual values 11. % Step 3: Solve Sub-Problem (SP) to generate new column 12. if compute_reduced_cost(dual_values) < 0 13. MP = update_MP(MP, generate_column()); % Add column if reduced cost is negative 14. end 15. % Step 4: Final Integer Solution (optional) 16. final_solution = branch_and_bound(MP); % Solve for an integer solution if needed 17. % Step 5: Display final solution 18. disp("Final solution: "); </pre>

5.2.2.2 Heuristic algorithm for Medical Resource Allocation

Non-dominated Sorting Genetic Algorithm II (NSGA-II) uses a Pareto sorting mechanism to balance multiple objectives and generate diverse, high-quality solutions [38]. It applies real-integer encoding to assign patients to surgery dates, ORs, and surgeons [39]. During decoding, the surgery sequence is optimized based on patient attributes (e.g., surgery duration and LOS) to minimize OR and surgeon overtime and reduce peak recovery bed demand, as shows in Fig. 6. This approach efficiently handles complex, multi-objective problems, and the complete process is outlined in pseudocode 4.

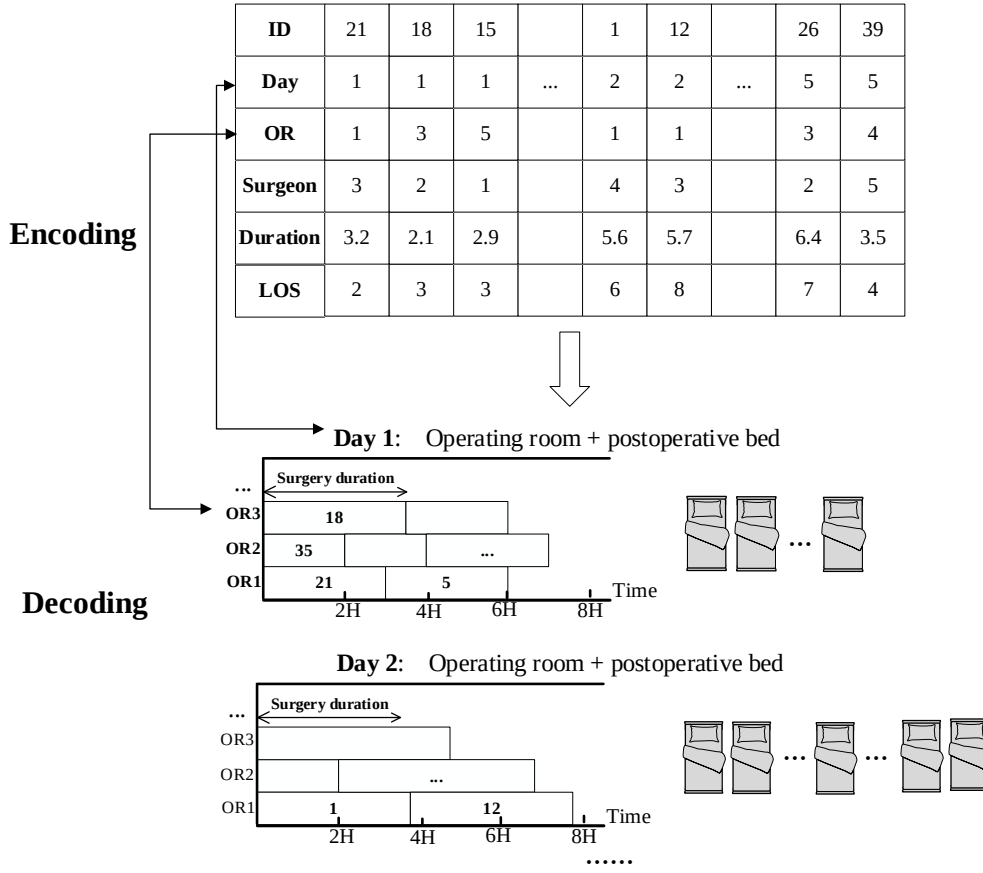


Fig. 6. Encoding and decoding method of NSGA-II algorithm

Pseudocode 4: NSGA-II for Resource Allocation Problem

1. % Step 1: Initialization of parameters
2. Initialize Population size, costs, working hours, etc.
3. population = initialize_population(population_size, ..., n_days, n_surgeons);
4. % Evaluate objectives for the initial population
5. Evaluate objectives f1, f2, f3 for population;
6. % Step 2: Non-dominated sorting and crowding distance
7. population = non_dominated_sorting(population); % Sort by Pareto ranking
8. % Main NSGA-II loop for each generation
9. for generation = 1:num_generations % Loop through generations
10. selected_parents = tournament_selection(population, population_size); % Parent selection
11. offspring = crossover(selected_parents); % Crossover operation
12. offspring = mutate(offspring); % Mutation operation
13. % Evaluate objectives for offspring
14. Evaluate objectives f1, f2, f3 for offspring;
15. % Combine population and offspring and Pareto ranking
16. combined_population = [population; offspring];
17. combined_population = non_dominated_sorting(combined_population);

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18. % Select the top individuals for the next generation
19.     population = select_top_N(combined_population, population_size);
20. end
21. % Step 5: Extract and display Pareto-optimal solutions
22. disp(pareto_solutions); % Display Pareto-optimal solutions

```

5.2.2.3 Results comparison

Detailed parameter settings for three comparative approaches are in the Table 4. The parameters were carefully chosen to align with the characteristics of the problem. Both Column Generation and NSGA-II were executed in MATLAB 2023b, with the results representing the best outcomes obtained after multiple runs.

Table 4 Parameter settings of the three approaches

Method	Parameter	Explanation	Value
ChatGPT-4o	Temperature	Balance creativity and determinism	0.75
	Max Tokens	Handle long and complex responses	2000
Column Generation	Initial Solution	Generate a starting solution	Greedy algorithm
	Stopping Criteria	Threshold for the duality gap	1e-4
	Subproblem Solver	Solve the subproblem optimally	Branch-and-Bound
	Objective Weight	Weights assigned for multi-objective	w1=0.01, w2=1, w3=1
NSGA-II	Population	Population Size	100
	Generations	Maximum number of iterations	200
	Mr	Crossover Rate	0.7
	Cr	Mutation rate	0.1
	Selection Method	Tournament selection	2

Table 5 shows the three objective functions and running times for ChatGPT-4o, Column Generation and NSGA-II. Fig. 7 illustrates the Gantt chart of OR allocation, the three objective functions, total OR overtime, and running times for these three methods.

Table 5. Objective functions and running times for three methods

Model	f1	f2	f3	Time(s)
ChatGPT-4o	25640	14.5	36	7.6
Column Generation	25280	5.8	34	478.6
NSGA-II	27760	22.2	37	15.7

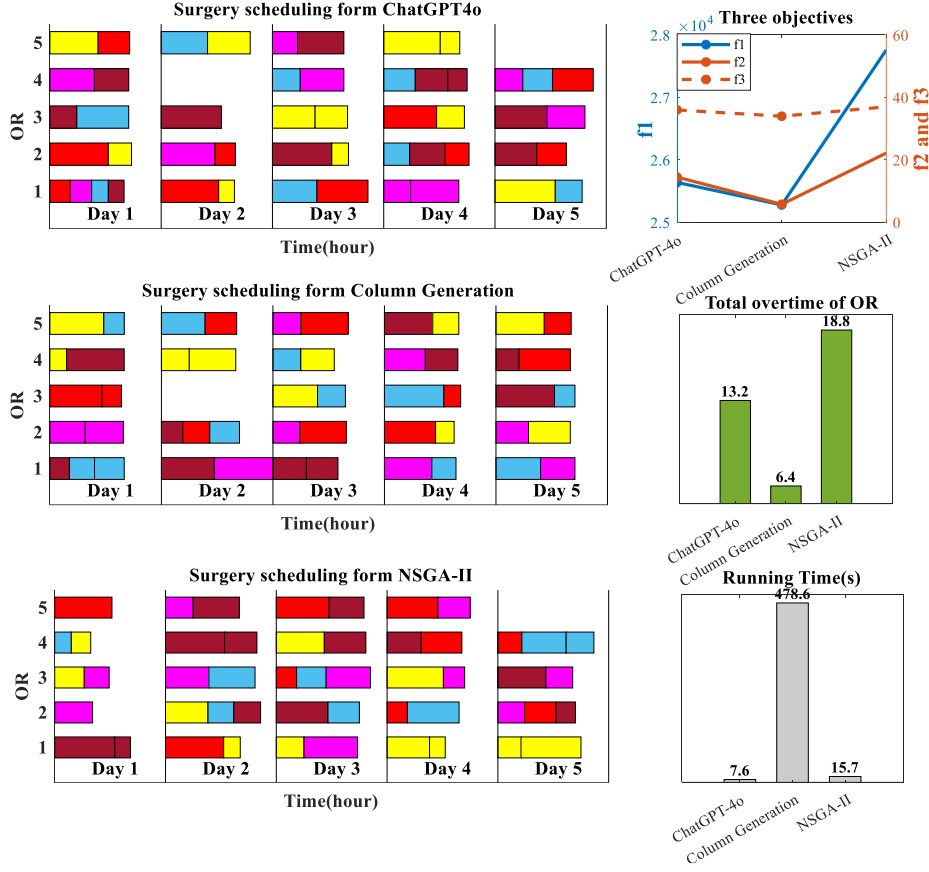


Fig. 7. OR allocation, total OR overtime, and running times for the three methods.

The Column Generation algorithm achieves optimal solutions but is challenging to apply in practice due to numerous constraints and the need to convert multi-objective problems into single-objective ones, requiring weight adjustments and scale unification. NSGA-II, while effective, demands careful encoding and decoding design, is prone to local optima, and requires tuning parameters like crossover and mutation rates, adding to its complexity. The powerful computational capabilities and adaptive nature of LLMs provide a new approach to solving complex multi-objective problems, with ChatGPT-4o delivering competitive solutions in the short time.

5.2.2.4 Statistical analysis

To analyze the statistical differences between the three algorithms and provide a robust comparison, we referred to the statistical methods outlined in [40], [41]. These methods emphasize the importance of using rigorous statistical tests to validate performance differences in optimization algorithms. Based on this, we applied a paired t-test to evaluate the performance of ChatGPT-4o against Column Generation and NSGA-II. Given the expectation that ChatGPT-4o outperforms the other two algorithms in terms of solution quality and runtime, a one-sided alternative hypothesis H1 as follows in Eq. (10).

$$H1: \mu_{ChatGPT-4o} - \mu_{OA} < 0 \quad (10)$$

where $\mu_{ChatGPT-4o}$ and μ_{OA} are the population means for ChatGPT-4o and OA (representing the compared algorithms: Column Generation and NSGA-II). The results of the

paired t-test are listed in Table 6.

Table 6. Results of paired t-test

Problem	ChatGPT-4o VS Column Generation	ChatGPT-4o VS NSGA-II
Mean Difference	-100.3	-516.8
P value	0.018	0.032

As shown in Table 6, for all three pairs, P was less than 0.1, and the mean difference is less than 0, which indicates that ChatGPT-4o is significantly different from other algorithms and performs better.

5.2.3. Comparison between problem size

Table 7 shows the results of allocating healthcare resources using seven LLMs, the Column Generation algorithm, and NSGA-II, as the patient scale increases from 50 to 55, 60, and 65 patients. The table includes the three objective functions, running times, and the number of patients left unallocated due to errors or omissions during the allocation process. Fig. 8 illustrates the corresponding results. For LLMs, when patient scales change, RAG dynamically incorporates new data, such as patient information and OR availability, enabling LLMs to generate updated allocation plans by combining prior experience with new inputs.

Table 7. Three objectives, running times, and patients left under different problem sizes

Num	Obj	Method								
		Chat GPT-3.5	LaMDA 2	PaLM 2	LLaMA 2	Qwen	Ernie	Chat GPT-4o	Column Generation	NSGA- II
55	f1	34780	31420	30060	32310	28990	31440	27780	27420	28160
	f2	44.1	31.9	41.1	48.8	42.4	37.2	37.1	11.9	31.1
	f3	55	49	41	47	46	50	40	36	42
	T(s)	14.3	9.4	8.2	11.1	11.3	9.8	7.3	566.4	20
	Left	0	0	13	2	4	0	0	0	0
60	f1	36060	32900	31580	33400	31940	32950	31220	30160	30420
	f2	56.2	59.1	55.2	56.3	53.5	60.9	41.5	29.2	43.8
	f3	58	53	46	51	50	54	43	39	46
	T(s)	12.9	8.7	8.1	11.6	11.9	9.4	6.2	601.5	25.5
	Left	0	2	15	5	7	0	0	0	0
65	f1	37260	36420	34600	35900	35990	37440	33940	33020	34980
	f2	58.2	62.2	52.5	62.4	54.8	65.1	46.4	40.3	52.4
	f3	62	55	49	55	53	57	46	39	48
	T(s)	13.4	8.9	7.9	10.8	11.4	8.6	6.8	782.3	30.4
	Left	0	3	17	7	10	0	0	0	0



Fig. 8. Three objectives, running times, and the number of patients left for 9 methods

For patient scales of 50, 55, 60, and 65, Column Generation consistently obtained the optimal solution, while ChatGPT-4o provided the second-best solution in the shortest time, and NSGA-II also achieved good results. However, exact algorithms take a long time, and as the problem size increases, the number of variables in the model rises sharply, causing optimization time to increase significantly.

In contrast, ChatGPT-4o and other LLMs maintained stable or slightly reduced solution times as problem size grew. Leveraging RAG, LLMs dynamically incorporate new patient data into scheduling, combining prior knowledge with new inputs to quickly generate updated plans without requiring complete re-optimization. This significantly reduces runtime, addresses the limitations of traditional methods, and advances LLM-driven dynamic resource allocation.

While NSGA-II produced satisfactory solutions, it required reloading data and re-optimizing for each problem size, resulting in longer runtimes. Some LLMs, like PaLM2 and Qwen, showed increasing errors as problem size grew, leaving more patients unallocated.

5.2.4 Comparison between the references

To demonstrate the effectiveness of the proposed method in solving multi-objective healthcare resource allocation problems, we compared it with several state-of-the-art approaches.

Liu et al. [42] proposed Evolution of Heuristics (EoH), a novel evolutionary paradigm that integrates LLMs with Evolutionary Computation methods for Automatic Heuristic Design. EoH leverages LLMs to generate heuristic strategies, which are then translated into executable code by the LLM itself. Similarly, Liu et al. [43] introduced LLMs as Evolutionary Optimizers (LMEA), where LLMs act as combinatorial optimizers by selecting parent solutions, performing crossover and mutation operations, and evaluating the offspring solutions for inclusion in the next generation. Cai et al. [44] developed the Multi-objective Evolutionary Algorithm based on Decomposition and further extended it with CoMOLS/D, a coevolutionary multi-objective local search framework.

These cutting-edge methods provide valuable benchmarks for evaluating the performance of our LLM-based approach in addressing multi-objective optimization problems. Table 8 shows the objectives and running times of Chatgpt4o, NSGA-II, EoH, LMEA, and CoMOLS/D on our four problem scales.

Table 8. The objectives and running times of five methods on four problem scales

Num	Obj	Method				
		ChatGPT-4o	EoH (Liu et al.[42])	LMEA (Liu et al.[43])	CoMOLS/D (Cai et al.[44])	NSGA-II
50	f1	25640	26220	26100	26480	27760
	f2	14.5	18.8	17.3	20.5	22.2
	f3	36	37	37	37	37
	T(s)	7.6	10.5	11.5	14.6	15.7
55	f1	27780	27900	27820	28060	28160
	f2	37.1	39.2	32.6	34.5	31.1
	f3	40	41	41	42	42
	T(s)	7.3	11.6	12.1	17.8	20
60	f1	31220	30840	31120	31460	30420
	f2	41.5	40.9	35	35.1	43.8
	f3	43	45	45	47	46
	T(s)	6.2	12.3	13.6	22.8	25.5
65	f1	33940	34120	33860	34700	34980
	f2	46.4	47.7	49.8	48.3	52.4
	f3	46	47	47	48	48
	T(s)	6.8	13.6	14.5	28.4	30.4

Figure 9 compares the five methods. EoH and LMEA show similar performance to ChatGPT-4o, as all three are based on LLMs. However, ChatGPT-4o achieves shorter runtime by avoiding the integration of traditional optimization techniques. CoMOLS/D and NSGA-II are also effective for multi-objective problems, with CoMOLS/D delivering better results and shorter runtimes. As the problem size grows, LLM-based methods outperform CoMOLS/D and NSGA-II, with runtime increases being less significant. This provides insights for dynamic resource allocation problems, as LLMs demonstrate minimal runtime increases and improved results as new data enlarges the problem scale.

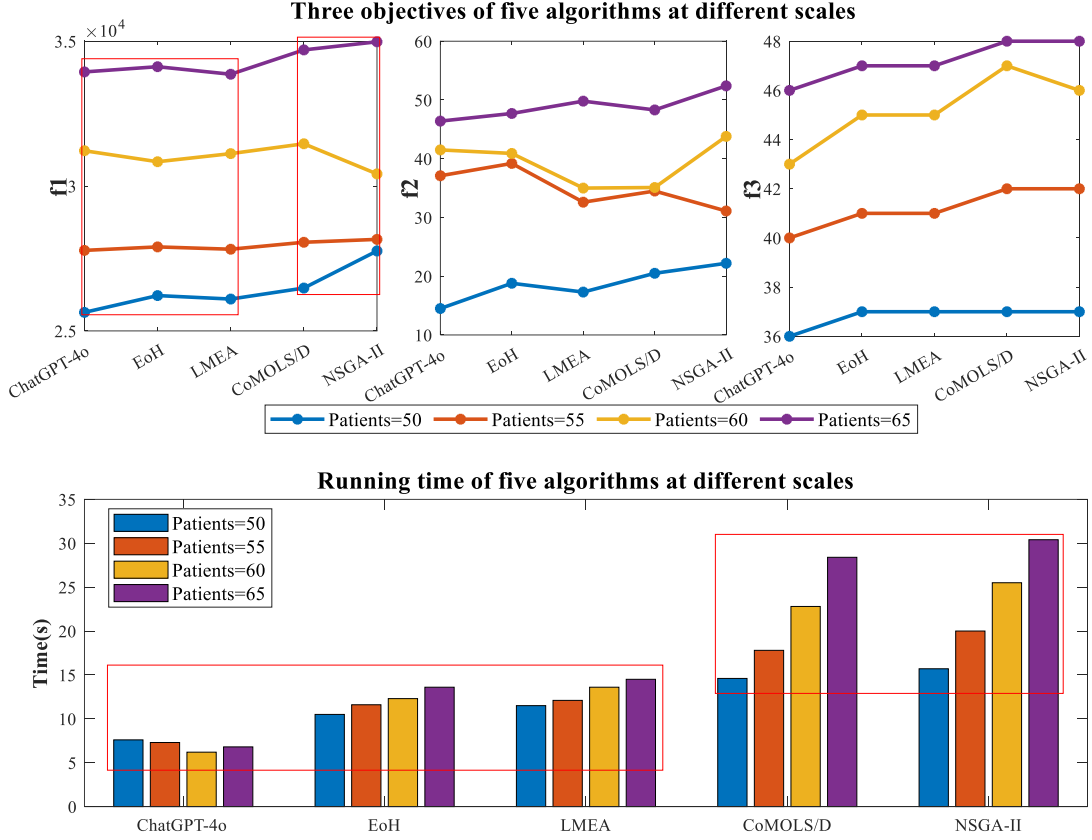


Fig. 9. The objectives and running times of five methods on different scales

6. Non-optimizable aspects and results

In addition to addressing the optimizable problems within the healthcare system, we also conducted research on non-model aspects, specifically focusing on medical knowledge Q&A. To better understand and analyze the factors influencing healthcare resource allocation, we designed a comprehensive questionnaire based on a survey of several medical institutions in Lyon, France. This questionnaire consists of 30 single-choice questions (2 points each) and 10 multiple-choice questions (4 points each), with a total score of 100 points. The questionnaire covers a wide range of topics including economics, humanities, ethics, and law, and involve key elements such as healthcare staff, patients, medical equipment, and hospitals.

To compare the performance of various LLMs in medical knowledge Q&A, we still selected above 7 LLMs models. Experts in the relevant fields evaluated and scored the answers provided by these models. Fig. 10 illustrates a process framework where LLMs solve Medical Knowledge Q&A.

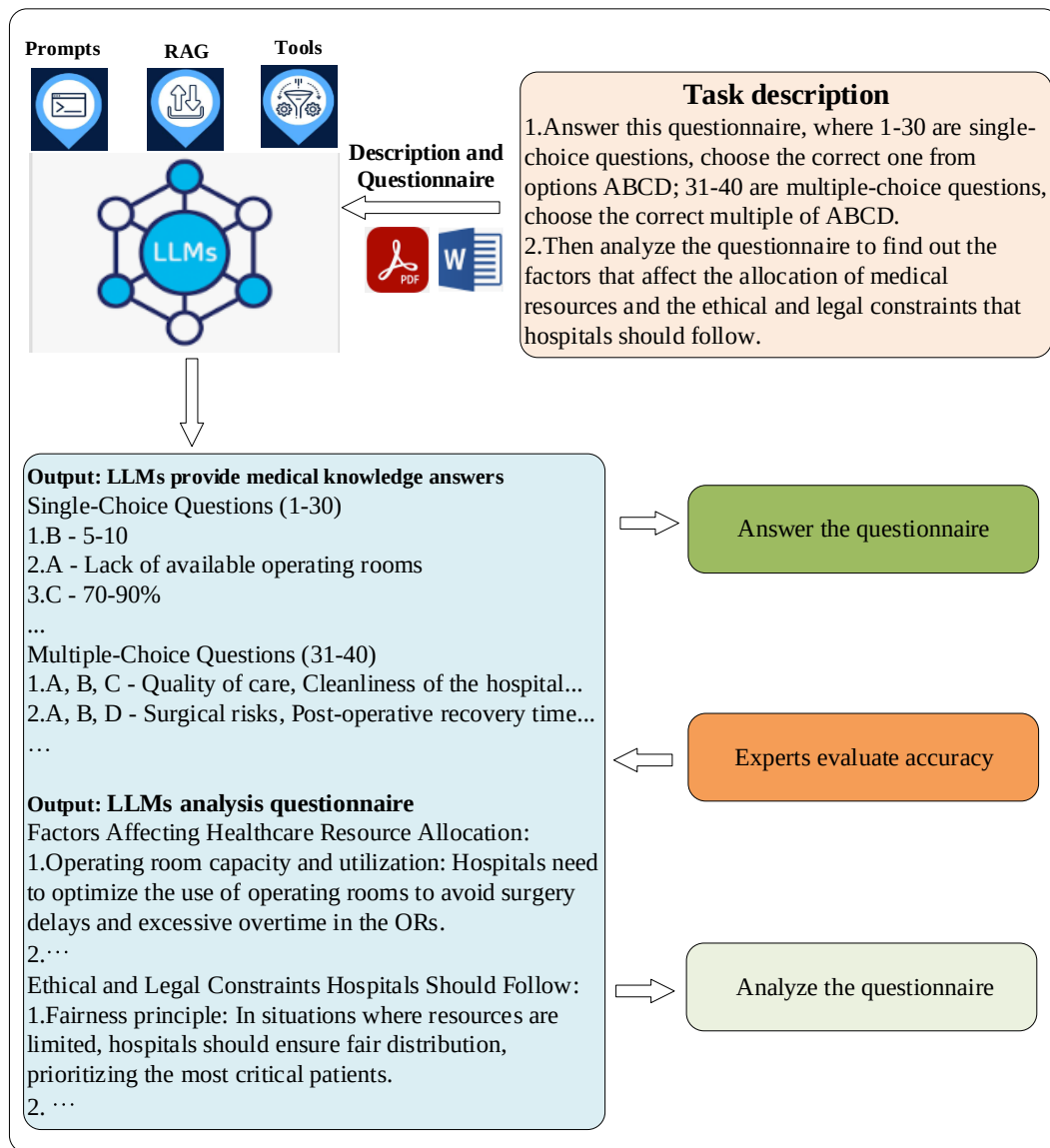


Fig. 10. The framework where LLMs solve Medical Knowledge Q&A.

Fig. 11 shows the number of correctly answered single-choice and multiple-choice questions for each model, along with the corresponding total score. The larger the bubble, the higher the total score. ChatGPT-4o and LaMDA 2 performed best on the single-choice questions, while Qwen also performed well on the multiple-choice questions. ChatGPT-3.5 had a higher error rate due to hallucination issues, which affected its accuracy. The ChatGPT-4o model achieved the highest accuracy and the shortest response time by enhancing domain-specific knowledge. The remaining five models, LaMDA 2, PaLM 2, LLaMA 2, Qwen, and Ernie, scored between 60 and 70 overall.

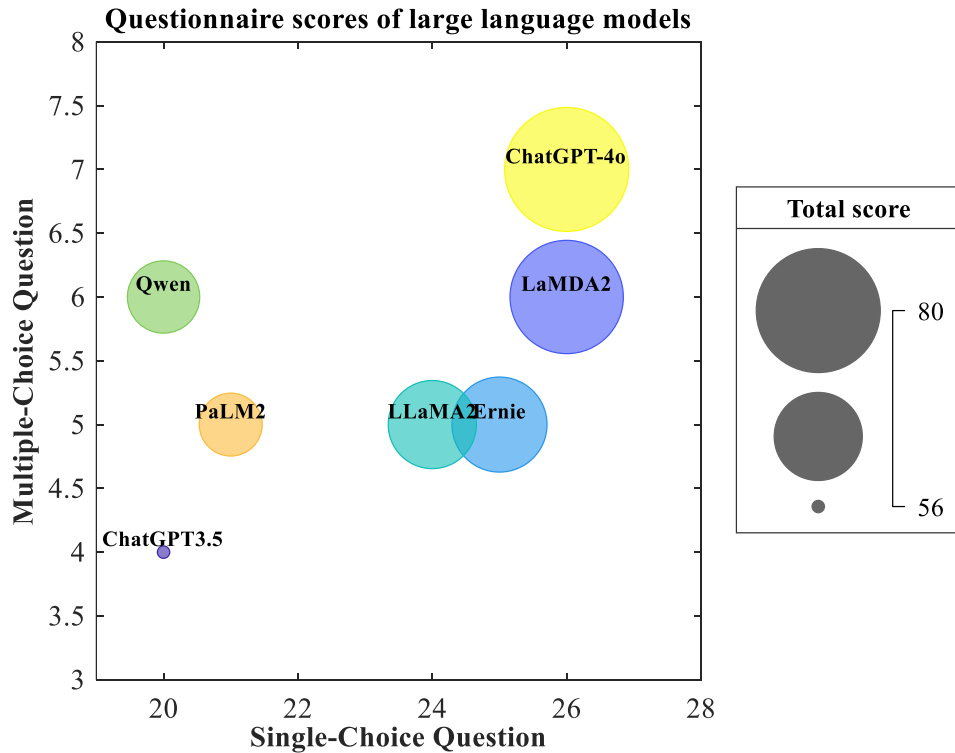


Fig. 11. The number of correct numbers and total scores

Based on the summary of these LLMs' analysis of this questionnaire, these are several critical factors that influence healthcare resource allocation. The capacity and utilization of ORs, which directly impact the scheduling of elective and emergency surgeries. Efficient management of recovery beds is also essential, as it ensures a smooth patient flow and prevents delays caused by bed shortages. Additionally, the availability of doctors and healthcare staff is vital, especially in high-demand areas like ORs and ICUs, where staff shortages can significantly affect resource allocation. Furthermore, the maintenance and availability of medical equipment play a pivotal role in surgical operations, as equipment issues can cause delays and increase operational inefficiencies. Hospitals must focus on optimizing these resources to balance patient care demands with resource constraints, ensuring smooth, uninterrupted operations.

Beyond the logistical challenges, healthcare resource allocation also involves significant legal and ethical complexities. Fairness in resource allocation is paramount, particularly in situations of limited resources, where prioritizing critical patients is essential to avoid inequities based on socioeconomic or other non-medical factors. Informed consent must be obtained from patients, ensuring they are fully aware of the risks, outcomes, and potential complications of their treatment. Patient privacy protection is another key consideration, requiring strict adherence to confidentiality laws when handling sensitive information. Additionally, transparency in the decision-making process is critical, especially during public health emergencies, to ensure that resource allocation is conducted in a fair, accountable manner. Above all, patient safety must be prioritized, with hospitals taking every measure to reduce complications and errors in care, while ensuring that resources are distributed in a way that maximizes patient outcomes and minimizes risks.

7. Conclusion

This study leverages large language models (LLMs) to address both optimizable and non-optimizable challenges in healthcare resource allocation. The LLMs used in this research incorporate three key techniques: Prompt Engineering, Retrieval-Augmented Generation (RAG), and Tool Utilization. Among the seven selected LLMs, including LaMDA 2, PaLM 2, and Qwen, ChatGPT-4o demonstrated outstanding performance by reducing overtime for ORs and surgeons and alleviating peak bed demand caused by patient congestion. However, models like PaLM 2, Qwen, and LLaMA 2 faced issues such as poor patient management and unbalanced OR utilization. ChatGPT-4o also achieved the highest accuracy in medical knowledge Q&A, indicating that healthcare resource allocation involves not only optimization but also complex legal and ethical considerations.

Compared to exact algorithms, heuristic methods, and other hybrid approaches, LLMs show great potential in solving healthcare resource optimization problems. LLMs can dynamically integrate medical data, adapting to changes in problem scale or demand without requiring re-optimization. This approach addresses the runtime limitations of traditional methods and advances LLM-driven dynamic resource allocation.

Despite these advantages, LLM-based methods also have limitations. Solution accuracy may be affected by the quality of input data, and outputs may exhibit randomness or hallucinations. Future research should focus on incorporating more complex factors, such as emergencies and uncertainties, to enhance the problem-solving capabilities and practical applicability of LLMs. Addressing these limitations may involve adopting ensemble techniques, reinforcement learning strategies, or hybrid approaches that combine LLMs with traditional optimization algorithms.

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